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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.5

$T_7\left(f,2\right)(x)$ voor $f(x) = \dfrac{1}{x-1}$

$f(x) = (x-1)^{-1}$

$f'(x) = -1(x-1)^{-2}$

$f''(x) = 2(x-1)^{-3}$

$f'''(x) = -6(x-1)^{-4}$

$f^{iv}(x) = 24(x-1)^{-5}$

$f^{v}(x) = -120(x-1)^{-6}$

$f^{vi}(x) = 720(x-1)^{-7}$

$f^{vii}(x) = -5040(x-1)^{-8}$

$f(2) = 1$

$f'(2) = -1$

$f''(2) = 2$

$f'''(2) = -6$

$f^{iv}(2) = 24$

$f^{v}(2) = -120$

$f^{vi}(2) = 720$

$f^{vii}(2) = -5040$

$$\Rightarrow T_7\left(f,2\right)(x) = \sum_{k=0}^{7} \frac{f^{(k)}(2)}{k!}(x-2)^k$$
$$= \frac{1}{0!} + \frac{-1}{1!}(x-2) + \frac{2}{2!}(x-2)^2 + \frac{-6}{3!}(x-2)^3 + \frac{24}{4!}(x-2)^4$$
$$+\frac{-120}{5!}(x-2)^5 + \frac{720}{6!}(x-2)^6 + \frac{-5040}{7!}(x-2)^7$$
$$= 1 - (x-2) + (x-2)^2 - (x-2)^3 + (x-2)^4 - (x-2)^5 + (x-2)^6 - (x-2)^7$$

References